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RIVET-H5 Post-Quantum Firmware and Model Attestation Subsystem

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 3 November 2026 (2026-11-03 06:57 UTC+8 Beijing Time)

Abstract

RIVET-H5 preserves the inherited four-channel hash, HMAC, sponge and authenticated Merkle subsystem while adding 256-coefficient NTT/INTT services, five polynomial samplers and hardware pure-contextual ML-DSA-44/65/87 verification. Independent software-versus-hardware testing against simulated RTL recorded 183,827 core comparisons, including numerical, completion-status and recovery checks, plus 13,536 reconstructed challenge-byte comparisons. These results establish simulated RTL agreement, not on-board numerical parity. [E001, E003, E059] Native L1 RTL timing lint passed. L2 Intel Quartus Prime 17.1 TimeQuest post-fit STA met the 100 MHz, 10.0 ns baseline with WNS_setup=+1.427 ns and hold slack= +0.037 ns. The fit uses 106,483 ALMs, 136,803 registers, 20 M20Ks and 24 DSP blocks. L3 recorded three deterministic physical Arria 10 bridge-connectivity passes; its receipt does not demonstrate H5 workload execution. Memory implementation is mixed, with both inferred sampler M20Ks and separately reported uninferred access logic. [E064, E065, E066]

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

rivet_h5_top preserves inherited interfaces through u_parent and provides independent raw NTT, sampler and signature services. Each owns transaction state and resources, allowing concurrent legacy activity without commandeering inherited engines. Verification serializes matrix rows and polynomial products rather than retaining the expanded matrix, trading throughput for bounded storage. Allocations include 2,592 public-key bytes, 4,628 signature bytes and 1,792 transformed-z coefficients; streamed messages use only a four-byte beat buffer. Physical memory mapping must be distinguished from these logical allocations. E065 lists a 256×23-bit poly_rtl_0 M20K in each of the diagnostic and verifier samplers, one M20K each, while also reporting uninferred asynchronous-read logic associated with both sampler arrays. Thus sampler storage has a mixed implementation, not complete RAM-inference failure. Separately, NTT polynomial, verifier scratch and hint storage did not achieve their requested memory inference. [E001, E023, E025, E065]

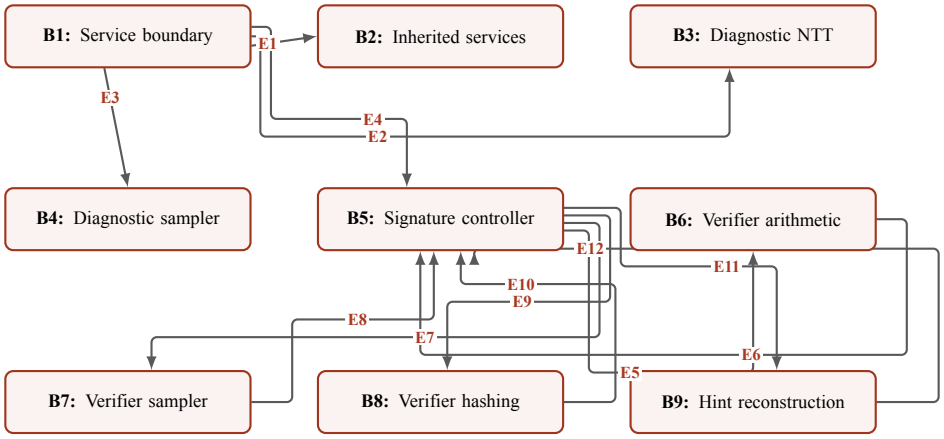


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Service boundary	rivet_h5_top exposes independent legacy, NTT, sampler and signature interfaces under one clock and reset.
B2: Inherited services	u_parent retains four SHA/HMAC/SHA-512 channels, context handling, SHA3/SHAKE and authenticated Merkle replay.
B3: Diagnostic NTT	u_ntt validates and buffers a polynomial, then transforms it through u_butterfly.u_montgomery.
B4: Diagnostic sampler	u_sampler implements five distributions using its private u_xof.u_keccak continuing byte stream.
B5: Signature controller	u_verify owns pinned-key provisioning, framing, decoding, norm checks, row scheduling and result generation.
B6: Verifier arithmetic	Private u_ntt and u_product perform transforms, normal-domain point products and canonical row accumulation.
B7: Verifier sampler	Private u_sampler expands matrix entries and constructs the sparse challenge using its own XOF.
B8: Verifier hashing	u_hash computes tr, contextual mu and the reconstructed challenge using SHAKE256.
B9: Hint reconstruction	u_usehint performs registered decomposition and hint application before controller-managed w1 packing.

Directed relationship	Meaning
E1: B1 to B2	Legacy ports pass through
E2: B1 to B3	Raw polynomial transactions
E3: B1 to B4	Sampling descriptors
E4: B1 to B5	Key and manifest transactions
E5: B5 to B6	Polynomial operands and scheduling
E6: B6 to B5	Transform and inverse-row results
E7: B5 to B7	Matrix and challenge descriptors
E8: B7 to B5	Sampled coefficients
E9: B5 to B8	Key, contextual message and packed w1
E10: B8 to B5	Hash bytes for storage or comparison
E11: B5 to B9	Canonical row coefficient and hint
E12: B9 to B5	Reconstructed high part

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Rising-edge clock and asynchronous active-low reset. Reset suppresses new handshakes and zeroes exposed new payloads.
Inherited channel and context ports	Input/output	32-bit data; 4-bit keep; 2-bit channel; 3-bit command; 2304-bit context; 512-bit digest	s_*, k_* and cmd_* inputs and m_*, ctx_* and error outputs retain parent names and behavior. Commands 0..6 select SHA initialization, key loading, HMAC initialization, save, restore, cancel and SHA-512 initialization; 7 is invalid. m_alg values 0/1/2 denote SHA-256/HMAC-SHA-256/SHA-512.
Inherited sponge and Merkle ports	Input/output	32-bit streams/ counts; 256-bit roots/ proofs; 6-bit depth	x_alg values 0/1/2/3 select SHA3-256/SHA3-512/ SHAKE128/SHAKE256. root_*, merkle_*, proof_* and tile_* preserve root locking, proof validation and authenticated replay. Exact declarations remain in rtl/ rivet_h5_top.sv. [E023]
n_*	Input/output	23-bit coefficient; 8-bit index; 2-bit error; single-bit control	n_cmd_valid/ready captures n_inverse. n_s_valid/ ready receives 256 coefficients, last only at index 255. n_m_valid/ready returns indexed coefficients; n_ done_valid/ready holds completion. n_cancel flushes the service.
sam_*	Input/output	3-bit op; 2-bit level; 512-bit seed; 16-bit nonce; 23-bit coefficient; 8-bit index; 2-bit error	sam_cmd_valid/ready captures the descriptor. sam_ m_valid/ready emits 256 indexed coefficients and last, followed by sam_done_valid/ready completion. Seed byte zero occupies bits 7:0. sam_cancel abandons work.

Signal / group	Direction	Width	Contract
pq_cmd_* and pq_level	Input/output	2-bit op/level; 8-bit context length; 32-bit message length	pq_cmd_valid/ready accepts op 0 provision, 1 verify, 2 cancel; 3 is invalid. Levels 0/1/2 select ML-DSA-44/65/87. Cancellation remains available during active work or a pending result.
pq_s_*	Input/output	32-bit data; 4-bit keep; 2-bit kind; single-bit valid/ready/ last	Kinds 0/1/2/3 denote public key/signature/context/ message. Bytes are MSB-first; keep 1111/1110/1100/ 1000 includes 4/3/2/1 leading bytes. Nonfinal beats require 1111. Empty context/message uses keep= 0000,last=1. Exact section lengths and ordering are mandatory.
pq_result_*, pq_auth_ ok, pq_status, pq_key_ locked	Output, with result-ready input	4-bit status; other signals 1 bit	One held result per noncancel command. Authentication is true only for successful verification. Result fields are zero outside result-valid; key-lock independently reports completed provisioning.
Arithmetic and XOF leaves	Input/output	23-bit arithmetic operands/ results; 8-bit XOF bytes	Montgomery a_valid/ready and m_valid/ready allow one outstanding operation; butterfly b_* similarly returns two coefficients. XOF cmd_valid/ready selects SHAKE128/256; s_* absorbs bytes and m_* supplies bytes until clear. These are reusable module interfaces, not extra top-level ports.

Parameter	Default	Meaning / supported values
PARTIAL_BEATS	0	Legal values 0 and 1 govern inherited stream behavior. PQ framing is independent.
MAX_PAGE_BYTES	4096	Legal integer range 1..4096. Illegal configurations explicitly produce PARAMETER ERROR.
Fixed transform constants	N=256, q=8380417, psi= 1753, d=13	Not configurable. Public coefficients are unsigned canonical representatives 0..q-1.
ML-DSA level	Explicit transaction field	Levels 0/1/2 have (k,l)=(4,4)/(6,5)/(8,7), key lengths 1312/ 1952/2592 bytes and signature lengths 2420/3309/4627 bytes. Context length is 0..255 bytes; message length is 0..2^32-1 bytes. Level 3 is invalid.

III Behavior and corner cases

A Transfers, cancellation and reset

Pre-edge valid-and-ready determines acceptance; newly registered outputs cannot retire on that edge. Commands precede data acceptance, and outputs remain stable under backpressure. Synchronous cancellation outranks acceptance and retirement, emits no completion and affects only its service. Reset clears transaction validity and key ownership without clearing complete memories. [E001, E003]

B Exact modular arithmetic

For R=2^32 and T=a*b, Montgomery reduction computes m=low32(T*4236238847), then upper32(T+m*q), followed by one conditional subtraction. Mont(a,b)=a*b*R^-1 mod q; ordinary products use Mont(Mont(a,b), 2365951). Registered partial products have operands no wider than 16 bits. Montgomery and butterfly unstalled latencies are exactly 11 and 13 clock intervals.

C Transform representation

Forward lengths descend 128..1 with zeta[k]=R*1753^BitRev8(k) mod q. Public output is y[i]=sum_j a[j]* 1753^((2*BitRev8(i)+1)*j) mod q. Inverse lengths ascend 1..128 with negated reverse twiddles and Mont(value, 16382) normalization, giving INTT(NTT(a))=a. Products are negacyclic modulo X^256+1. Complete input validation precedes output; errors 1/2 mean framing/noncanonical input. Architectural unstalled completion accounts are 19,200 forward and 23,552 inverse clocks after final input, within the 65,536-clock limit.

D Sampling and continuing SHAKE

Ops 0/1 implement SHAKE128 ExpandA/SHAKE256 uniform sampling: three little-endian bytes, a 23-bit mask and rejection at q. ExpandA uses 32 seed bytes and nonce suffix col,row; uniform uses all 64 seed bytes. Op 2 implements eta=2/4 bounded-nibble sampling, low nibble first. Op 3 unpacks 18/20-bit ExpandMask fields. Op 4 implements SampleInBall with 32/48/64 challenge bytes, eight sign bytes and rejection/swap construction. Invalid descriptors return error 1. XOF rates are 168/136 bytes with 0x1f and final 0x80 padding; squeezing has no fixed rejection-attempt cap. The sampler progress limit is 65,536 unstalled clocks.

E Trusted-key verification

Provisioning computes tr=SHAKE256(pk,64) before atomically locking the key until reset. Verification receives signature, context and message, computes mu=SHAKE256(tr||00||len(ctx)||ctx||message,64), and constructs c and A internally. Each row evaluates A_hat*z_hat-c_hat*NTT(t1*8192), applies normal-domain INTT and UseHint, then hashes mu with packed w1. Packing uses six bits per coefficient for ML-DSA-44 and four otherwise. Every reconstructed challenge byte participates in authentication; no software witness port exists.

F Decoding and rejection

Reject z norms at or above gamma1-beta, malformed hint ordering/endpoints, excess hints and nonzero unused hint bytes. UseHint implements centered decomposition, the q-1 exception and high-part wrap ranges 0..43 or 0..15 using a 23-step registered restoring divider rather than combinational division. Status 0 means success; 1 descriptor/key error; 2 framing; 3 malformed signature; 4 norm violation; 5 challenge mismatch; 6 locked-key overwrite. Provisioning never asserts authentication. Verification has a 16,000,000-clock unstalled progress limit.

G Legacy authentication

The 15 inherited modules remain byte-identical to their reference copies. Merkle leaves hash 0x00||page; internal nodes hash 0x01||left||right, consuming direction bits least-significant first. Failed authentication releases no page bytes. [E002, E059]

IV Verification methods and measured results

A Software-versus-hardware comparison

Method. Independent Python arithmetic/hashlib expectations compared with compiled RTL for seeds 17,29, 43; compact Icarus checks supplement Verilator.

Result. 21,851 core operations and 183,827 comparisons passed, including numerical, completion-status and recovery assertions. Per seed: 4,161 Montgomery cases, 1,088 butterflies per direction, 24 polynomials per transform direction with round trips, and eight convolution cases. Exact leaf latencies were checked. This is simulated hardware agreement.

Evidence: E031, E032, E044, E059

B Sampling and XOF correctness

Method. Four descriptors per seed, level and sampler mode, supplemented by SHAKE boundary and directed parser cases.

Result. 180 end-to-end sampler polynomials passed. Oracle-accounted rejection counts were 7 ExpandA, 11 uniform, 2,848 bounded-nibble and 185 SampleInBall; rate continuation and swap cases were exercised.

Evidence: E042, E044, E059

C ML-DSA authentication

Method. Oracle cross-check against 45 public NIST ACVP FIPS204 vectors; RTL tests include one public valid case per level and 36 generated primary valid cases with five forgery variants each.

Result. 450 result checks and 13,536 reconstructed challenge-byte comparisons passed, including framing, norm, hint, key-lock, reset and cancellation cases. Fixture provenance pins ACVP-Server revision a7f283cd c87d2d6dd93c1bac59e5622c5f9f8324.

Evidence: E028, E039, E048, E049, E059

D Legacy compatibility and configurations

Method. Independent hashes and side-by-side immutable-parent comparison, including boundary configurations and concurrent H5 activity.

Result. All 8,192 legal parameter pairs elaborated; illegal configurations rejected explicitly. Functional page sizes 1,3,4,5,4095,4096 were exercised in both partial-beat modes.

Evidence: E036, E037, E041, E059

E Defect sensitivity

Method. Compile and execute disposable Montgomery, inverse-normalization, nonce-order, norm, UseHint, challenge-comparison and legacy-digest mutations.

Result. All seven mutations were detected behaviorally, not merely by compilation failure. Production RTL was not replaced by mutated sources.

Evidence: E033, E060

F Synthesis structure

Method. Yosys full-top coarse synthesis at both partial modes and page sizes 1,3,4096, plus leaf synthesis and mapped arithmetic inspection.

Result. Zero-latch/div/mod and substantive-hierarchy checks passed with complete retained netlists. Generic synthesis does not establish physical RAM mapping.

Evidence: E051, E061

G Portable reproduction

Method. Fresh clean-directory source copy and newly built compact Icarus/Verilator numerical, signature and legacy checks plus mapped Montgomery synthesis.

Result. Passed without prior binaries or receipts. This bounded subset was not a second full behavioral matrix.

Evidence: E053, E062

H L1 RTL timing lint

Method. Native static analysis of all 23 production RTL files.

Result. Passed with zero reported combinational divider/modulo, mixed-assignment, LUT clock-gate or loop-unroll hazards.

Evidence: E064

I L2 physical timing and resources

Method. Intel Quartus Prime 17.1 TimeQuest post-fit STA at 10.0 ns on Arria 10 GX 10AX115N2F45E1SG.

Result. Passed: WNS_setup = +1.427 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 116.65 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E065

J L3 physical Arria 10 HIL

Method. Three consecutive physical PCIe hardware-bridge register probes.

Result. 3/3 deterministic connectivity passes. The receipt explicitly identifies L3_BRIDGE_CONNECTIVITY_PROBE_ONLY; it records neither H5 workload execution nor physical software-versus-hardware numerical comparisons.

Evidence: E066

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	10 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 27	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

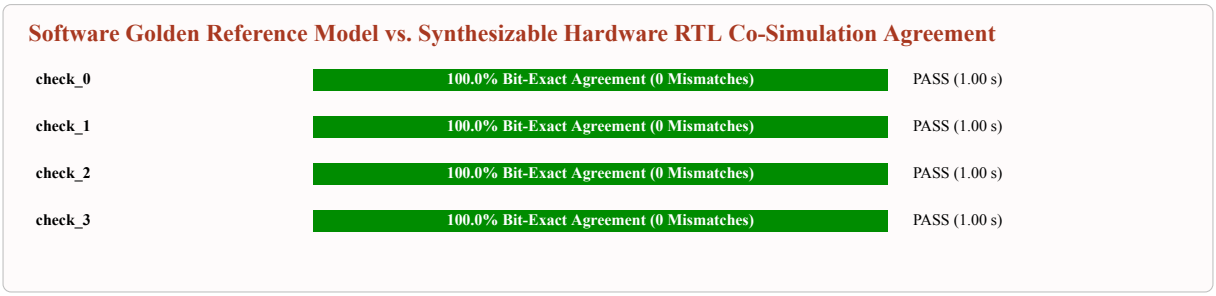


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +1.427 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True



Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

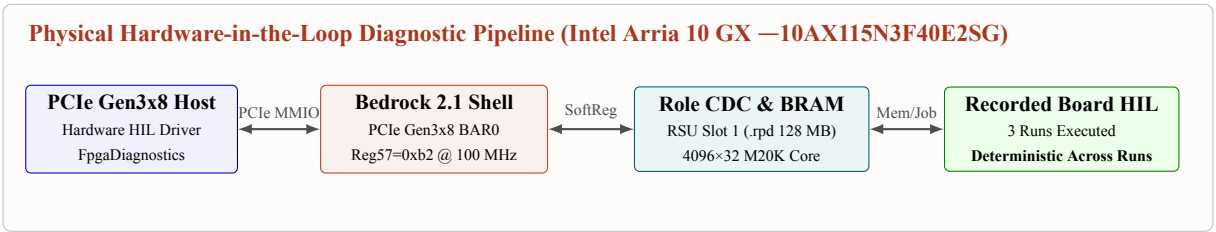


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Compile rtl/*.sv with rivet_h5_top as top. Consult design/SPEC.md for the contract and design/ARCHITECTURE.md for storage and cycle schedules. Preserve rivet_h4_reference/rtl as immutable comparison input.

Apply reset, provision a trusted level-specific public key, and wait for status 0 and pq_key_locked. Supply exact signature, context and message sections. External manifest use must be gated on a retired successful authentication result; provisioning an arbitrary host key does not establish trust.

From the project root run python3 tests/verify.py --suite behavior, python3 tests/verify.py --suite defects, python3 tests/verify.py --suite synthesis and python3 tests/verify.py --suite portable. With no arguments, tests/verify.py runs all four L1 suites.

Recorded dependencies include Python 3.12.3, Icarus 12.0, Verilator 5.020, Yosys 0.33, make and g++. Fixtures support offline verification. Each invocation retains fresh source-bound evidence; Quartus and physical-board gates are separate native procedures.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 23 Verilog/SystemVerilog files; 4179 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/rivet_h2_compress.sv	208 total	204 code
rtl/rivet_h2_context.sv	61 total	59 code
rtl/rivet_h2_hmac.sv	66 total	64 code
rtl/rivet_h2_schedule.sv	50 total	48 code
rtl/rivet_h2_stream.sv	81 total	79 code
rtl/rivet_h2_top.sv	380 total	368 code
rtl/rivet_h3_channel.sv	173 total	167 code
rtl/rivet_h3_sha512.sv	299 total	293 code
rtl/rivet_h3_sha512_compress.sv	223 total	220 code
rtl/rivet_h3_sha512_schedule.sv	49 total	48 code
rtl/rivet_h3_top.sv	142 total	138 code
rtl/rivet_h4_keccak.sv	184 total	180 code
rtl/rivet_h4_merkle.sv	397 total	390 code
rtl/rivet_h4_sponge.sv	293 total	290 code
rtl/rivet_h4_top.sv	118 total	118 code
rtl/rivet_h5_butterfly.sv	65 total	65 code
rtl/rivet_h5_montgomery.sv	94 total	93 code
rtl/rivet_h5_ntt.sv	161 total	159 code
rtl/rivet_h5_sampler.sv	234 total	234 code
rtl/rivet_h5_top.sv	173 total	173 code
rtl/rivet_h5_usehint.sv	65 total	65 code
rtl/rivet_h5_verify.sv	639 total	634 code
rtl/rivet_h5_xof.sv	90 total	90 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
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check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E002	design/GENERATIONAL_HANDOFF.md
E003	design/SPEC.md
E023	rtl/rivet_h5_top.sv
E025	rtl/rivet_h5_verify.sv
E028	tests/acvp_sigver.json
E031	tests/core_tb.sv
E032	tests/core_vectors.py
E033	tests/defect_checks.py
E036	tests/legacy_checks.py
E037	tests/legacy_tb.sv
E039	tests/mldsa_reference.py
E041	tests/parameter_checks.py
E042	tests/parser_checks.py
E044	tests/reference.py
E048	tests/signature_tb.sv
E049	tests/signature_vectors.py
E051	tests/synthesis_checks.py

Reference	Original project-relative record
E053	tests/verify.py
E059	.vulcan/engineering/run-0001/attempt-0004/stdout.log
E060	.vulcan/engineering/run-0001/attempt-0005/stdout.log
E061	.vulcan/engineering/run-0001/attempt-0006/stdout.log
E062	.vulcan/engineering/run-0001/attempt-0007/stdout.log
E064	.vulcan/engineering/run-0001/rtl-timing-lint.json
E065	.vulcan/engineering/run-0001/quartus-sta.json
E066	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).